TaxaFlag

Post-assignment quality flagging for the TaxaID ecosystem. Identifies anomalous detections in taxonomic assignment results using data-driven and expert-review approaches.

Taxonomic assignment matches observations to identifications, but samples often contain artifacts. TaxaFlag provides three independent post-hoc checks:

- **Contamination screening** compares taxon read counts against control samples (lab blanks, field blanks, positive controls). For example, in metabarcoding, human and food-related sequences commonly appear in blanks, and their relative frequency in controls can justify removal.
- **Handler artifact detection** flags observations that fall within a time buffer around camera setup and retrieval, when human activity is expected.
- **LLM expert review** uses a language model to assess whether each assignment is plausible given the habitat and geographic location, flagging unusual detections for closer inspection.

Flagging Methods

Method	Function	Data needed
Contamination	`flag_contaminant()`	Read counts + control samples
Handler artifacts	`flag_handler()`	Timestamps + setup/retrieval times
Expert review	`review_assignments()`	Assignments + LLM API key

Each method is independent – use one, two, or all three depending on your data type and available metadata. Flags are additive columns, not filters. Together, these checks target false positives that survive upstream statistical assignment: contamination, handler artifacts, allochthonous transport (e.g., eDNA carried by currents from outside the sampling area), and other ecologically implausible detections.

Methods

Contamination Scoring

`flag_contaminant()` compares the relative abundance of each taxon in field samples versus control samples. Within each sample, read counts are first converted to proportions (reads for taxon / total reads), normalizing for sequencing depth. Proportions are then averaged across field and control replicates, and a score is computed:

```
score = mean_prop_field / (mean_prop_field + mean_prop_control)
```

Scores range from 0 (taxon found only in controls) to 1 (taxon found only in field samples). Taxa absent from controls receive a score of 1.0; taxa absent from field samples receive 0.0. Default thresholds classify scores as `"high"` risk (score 0.5, probable contaminant), `"moderate"` risk (0.5 < score 0.9, ambiguous), or `"low"` risk (score > 0.9, likely genuine detection). For positive controls, the interpretation inverts: taxa from positive controls appearing in field samples indicate cross-contamination.

Handler Artifact Detection

`flag_handler()` identifies observations that fall within a time buffer of camera setup or retrieval, when human activity is expected. For each group (e.g., camera station), the function identifies the earliest and latest timestamps as the sampling-period edges. Each observation receives a linear score based on its proximity to the nearest edge:

```
handler_score = minutes_to_nearest_edge / interval_minutes
```

clamped to [0, 1]. The default interval is 30 minutes. Observations outside the interval score 1.0 (valid); those at the exact edge score 0.0 (probable artifact). When `handler_taxa` is specified (e.g., "Homo sapiens"), only those taxa are scored for temporal proximity – other species detected near edges are assumed legitimate.

LLM Expert Review

`review_assignments()` submits each unique taxon (in batches of 15) to an LLM for structured assessment across eight dimensions: habitat fit, geographic plausibility, taxonomic scope, contamination risk, plausible alternatives, finer-rank hypotheses, confidence, and a free-text comment. Plausibility columns use a consistent vocabulary: `"likely"` / `"possible"` / `"unlikely"` (higher = more plausible genuine detection). Contamination risk uses `"low"` / `"moderate"` / `"high"` (higher = more contamination risk). The function includes truncation recovery: if an LLM response is cut off mid-JSON, it walks backward to find the last complete object and parses what is available. Taxa omitted by the LLM are filled with NA. Supports eDNA, acoustic, and image data via the `data_type` param. This review is intended as a structured second opinion, not an automated filter – users should treat the flags as candidates for closer inspection.

Installation

```r
# Requires TaxaTools (foundation package)
devtools::install("path/to/TaxaTools")
devtools::install("path/to/TaxaFlag")
```

Quick Start

Flag contamination from lab blanks

```r
library(TaxaFlag)

flagged <- flag_contaminant(
  df               = reads_long,
  taxon_col        = "taxon_name",
  reads_col        = "n_reads",
  event_col        = "event_id",
  control_samples  = c("Blank_1", "Blank_2"),
  contaminant_type = "lab_contaminant"
)

# Output adds: lab_contaminant_risk, lab_contaminant_score, lab_contaminant_reason
# Filter to high-risk taxa (probable contaminants)
flagged[flagged$lab_contaminant_risk == "high", ]
```

Flag handler artifacts (camera traps)

```
flagged <- flag_handler(
  df               = camera_detections,
  datetime_col     = "datetime",
  group_col        = "camera_id",
  interval_minutes = 30
)
```

LLM expert review

```
reviewed <- review_assignments(
  df           = consensus_results,
  taxon_col    = "consensus_taxon",
  context      = list(geography = "Southern California", habitat = "Marine"),
  target_group = "fish"
)
# Returns 8 structured columns:
# habitat_plausibility, geographic_plausibility, scope_plausibility,
# contamination_risk, review_alternatives, review_lower_hypotheses,
# review_confidence, review_comment

# Filter to contamination concerns
reviewed[reviewed$contamination_risk %in% c("high", "moderate"), ]
```

Reporting

```
# Generate a report section for assemble_report()
section <- report_flags(flagged)
```

Vignettes

- Quality Flagging – full workflow guide

Part of TaxaID

TaxaFlag is the final step in the TaxaID pipeline. It receives assignments from TaxaAssign and produces quality-annotated output for interpretation.

Ecosystem: TaxaAssign -> **TaxaFlag**

See the TaxaID README for ecosystem overview and installation instructions.

Citation

Lafferty, K.D., 2026, TaxaID – A modular R ecosystem for Bayesian taxonomic assignment: U.S. Geological Survey software release, https://doi.org/10.5066/xxxxxx.

Software Requirements

- R (>= 4.1.0)
- TaxaTools (for LLM provider functions and report assembly)
- An LLM API key is needed for `review_assignments()`

All dependencies are declared in the DESCRIPTION file and installed automatically.

Developed with Claude Code (Anthropic).